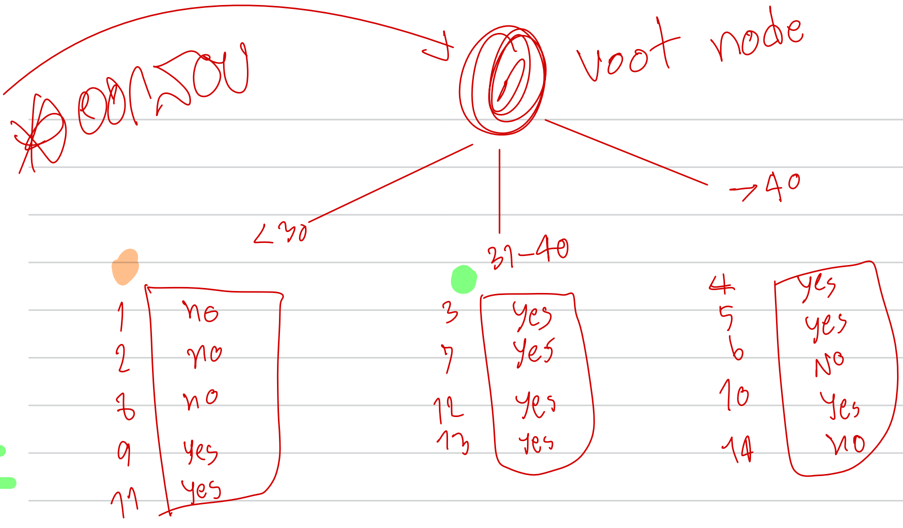
Training data set: Who buys computer?

	age	income	student	credit_rating	buys_computer
1	<=30	high	no	fair	no
2	<=30	high	no	excellent	no
	31…40	high	no	fair	yes
	>40	medium	no	fair	yes
	>40	low	yes	fair	yes
	>40	low	yes	excellent	no
	31…40	low	yes	excellent	yes
	<=30	medium	no	fair	no
	<=30	low	yes	fair	yes
	>40	medium	yes	fair	yes
	<=30	medium	yes	excellent	yes
	31…40	medium	no	excellent	yes
	31…40	high	yes	fair	yes
	>40	medium	no	excellent	no

y

Note: The data set is adapted from "Playing Tennis" example of R. Quinlan



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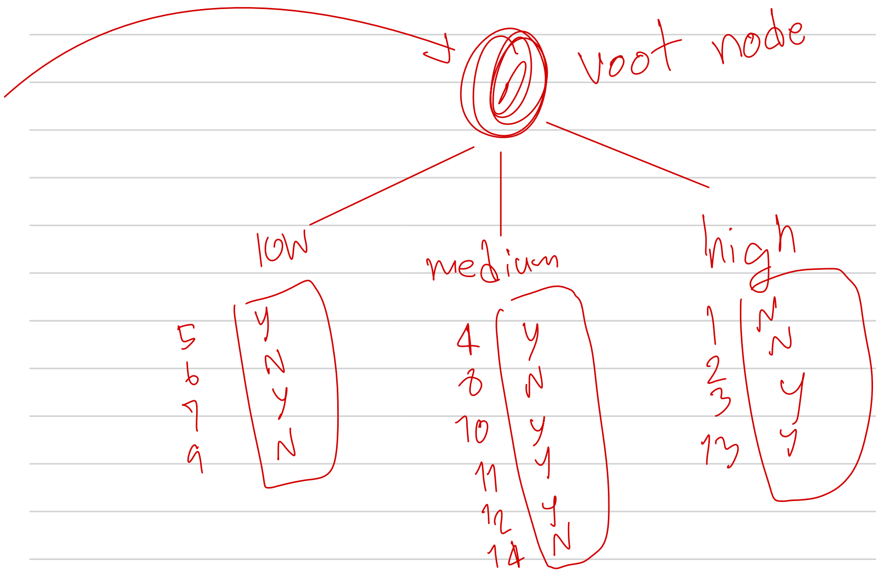
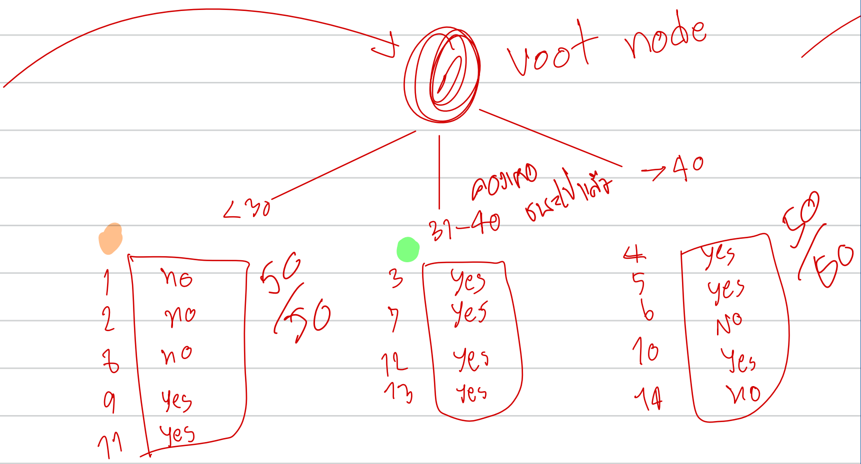
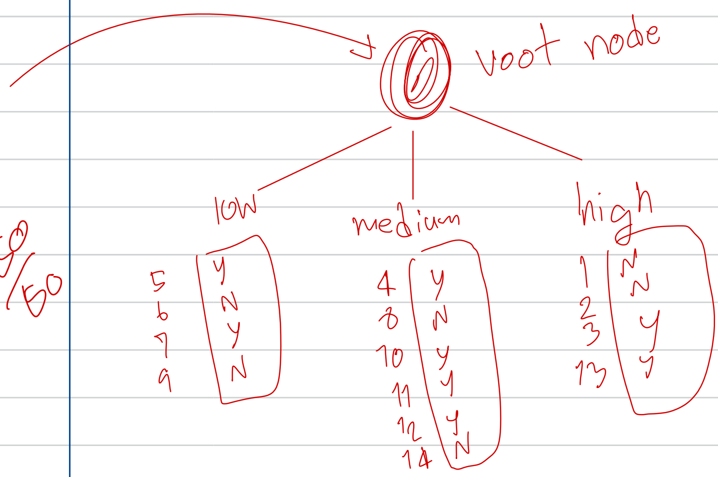




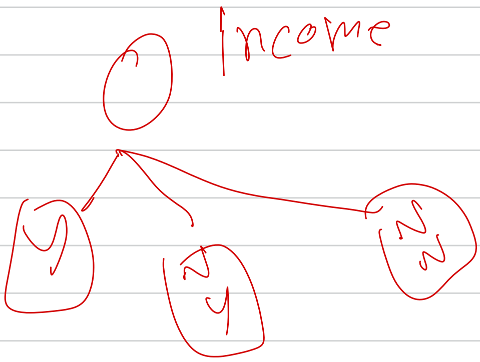


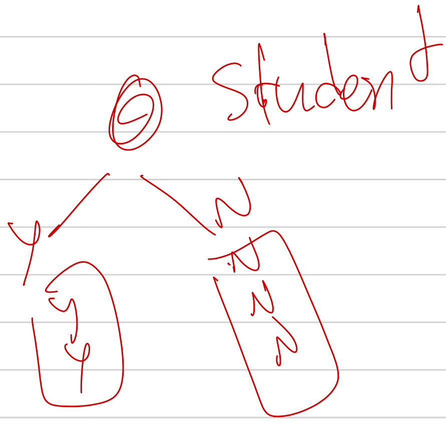
Age ← เป็นโหนดแรก

1	no
2	no
3	no
9	yes
11	yes

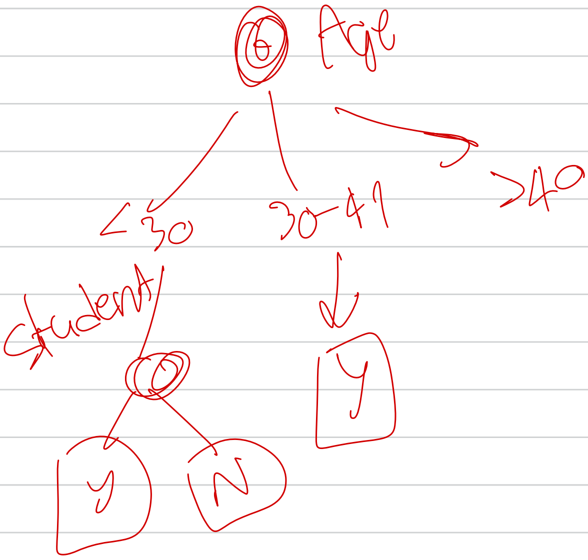
age	income	student	credit_rating	buys_computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31…40	high	no	fair	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
<=30	medium	yes	excellent	yes

ใช้แล้ว





หลัก



สำหรับกรณีข้อมูลไม่เป็นสมบูรณ์

con que

พื้น

Y
y
N

⇒ ดูจำนวน มากสุด

66.6%

ของเป็น

1.

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<=30	low	yes	fair	yes
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<=30	medium	yes	excellent	yes
31…40	medium	no	excellent	yes
31…40	high	yes	fair	yes
>40	medium	no	excellent	no

y

Note: The data set is adapted from "Playing Tennis" example of R. Quinlan

Student

Yes	No
y	N
N	N
y	y
y	y
y	N
y	y
y	N

Credit_rating

Fair	Excellent
N	N
y	N
y	y
y	y
N	y
y	N
y	
y	

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31…40	medium	no	excellent	yes
31…40	high	yes	fair	yes
>40	medium	no	excellent	no

y

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Income

$= \frac{4}{14} I(3,1) + \frac{6}{14} I(4,2) + \frac{4}{14} I(2,2)$

$= 0.232 + 0.394 + 0.285$

$= 0.911$

$Gain_{income} = 0.940 - 0.911 = 0.029$

Student

$= \frac{7}{14} I(6,1) + \frac{7}{14} I(3,4)$

$= 0.296 - 0.493$

$= 0.789$

$\therefore Gain_{student} = 0.940 - 0.789 = 0.151$

Credit_rating

$= \frac{8}{14} I(6,2) + \frac{6}{14} I(3,3)$

$= 0.464 + 0.429$

$= 0.893$

$Gian_{credit\_rating} = 0.940 - 0.893 = 0.047$

ก่อนจัด − หลังจัด

age $\quad Gain(A) = info(D) - Info_{age}(D)$

ก่อนจัด $Info(D)$

$= I(9,5) = -\frac{9}{14}\log_2\left(\frac{9}{14}\right) - \left(\frac{5}{14}\right)\log_2\left(\frac{5}{14}\right) = 0.94$

หลังจัด $info_{age}(D)$

$= \frac{5}{14} I(2,3) + \frac{4}{14} I(4,0) + \frac{5}{14} I(3,2)$

$= \frac{5}{14}\left[-\frac{2}{5}\log_2\left(\frac{2}{5}\right) - \frac{3}{5}\log_2\left(\frac{3}{5}\right)\right] + \cancel{\frac{4}{14} I(4,0)}^{0} + \frac{5}{14}\left[-\frac{3}{5}\log_2\left(\frac{3}{5}\right) - \frac{2}{5}\log_2\left(\frac{2}{5}\right)\right]$

$= 0.694$

ได้ว่า $\leq 30 \; I(2,3) = 0.971$

$30-40 \; (4,0) = 0$

$>40 \; (3,2) = 0.971$

$Gain_{age} = 0.940 - 0.694 = 0.246$

6630205 86 − 4

เสฏฐวุฒิ นาคสังข์